

SOLUTIONS_LEVEL2

SOLUTIONS LEVEL 2

BASIC MCQs – 200 QUESTIONS

Pattern: Single-correct MCQs (NCERT + JEE Main basic)

SECTION 1: BASIC CONCEPTS & TYPES OF SOLUTIONS (Q1–Q25)

1. A solution is best described as
 - A. A pure compound
 - B. A homogeneous mixture
 - C. A heterogeneous mixture
 - D. An element
2. A binary solution consists of
 - A. One solute and one solvent only
 - B. Only one component
 - C. Two or more solutes in one solvent
 - D. Only gases
3. In a solution of sugar in water, sugar acts as
 - A. Solvent
 - B. Solute
 - C. Co-solvent
 - D. Suspension
4. Brass (Cu–Zn alloy) is an example of
 - A. Solid in liquid solution
 - B. Liquid in liquid solution
 - C. Solid in solid solution
 - D. Gas in solid solution
5. Air is an example of
 - A. Gas in gas solution
 - B. Gas in liquid solution

- C. Liquid in gas solution
 - D. Gas in solid solution
6. Which of the following is an example of liquid in liquid solution?
- A. Alcohol in water
 - B. Sugar in water
 - C. Oxygen in water
 - D. Hydrogen in palladium
7. Hydrogen in palladium is an example of
- A. Solid in solid solution
 - B. Gas in solid solution
 - C. Gas in gas solution
 - D. Solid in gas solution
8. A saturated solution at a given temperature is one which
- A. Contains no solute
 - B. Contains maximum amount of solute in equilibrium with undissolved solute
 - C. Cannot dissolve any solute at any temperature
 - D. Contains less solute than required
9. An unsaturated solution is one which
- A. Contains no solute
 - B. Contains more solute than solubility limit
 - C. Contains less solute than solubility limit
 - D. Contains only solid solute
10. A supersaturated solution is
- A. Always unstable
 - B. Always stable
 - C. Exactly at solubility limit
 - D. A heterogeneous mixture
11. Which of the following is an aqueous solution?
- A. NaCl in benzene
 - B. Sugar in water
 - C. Iodine in CCl₄
 - D. Naphthalene in benzene

12. Which is a non-aqueous solution?
- A. NaCl in water
 - B. Glucose in water
 - C. Iodine in ethanol
 - D. CO₂ in water
13. The solute is the component which is
- A. In smaller amount by moles
 - B. In larger amount by mass
 - C. Always liquid
 - D. Always solid
14. The solvent is the component which
- A. Determines the physical state of solution
 - B. Is always water
 - C. Is always volatile
 - D. Is present in least amount
15. A solution of two miscible liquids is generally
- A. Heterogeneous
 - B. Homogeneous
 - C. Colloidal
 - D. Suspension
16. The number of components in a binary solution is
- A. 1
 - B. 2
 - C. 3
 - D. 4
17. A solution of CO₂ in water (soft drink) is
- A. Gas in gas
 - B. Gas in liquid
 - C. Liquid in gas
 - D. Solid in gas
18. Which of the following is NOT a correct pair?
- A. Brass – solid in solid

- B. Sugar in water – solid in liquid
 - C. Air – gas in gas
 - D. Iodine in water – liquid in liquid
19. Which of the following is always homogeneous?
- A. Colloid
 - B. Suspension
 - C. True solution
 - D. Emulsion
20. Colloidal solutions differ from true solutions mainly in
- A. Nature of solvent
 - B. Size of solute particles
 - C. Number of components
 - D. Being always solid
21. The term “solute” is best defined as
- A. The volatile component in the mixture
 - B. The component present in smaller amount
 - C. The component with lowest boiling point
 - D. The ionic part only
22. If both components are in equal amounts, the solvent is usually taken as
- A. The more volatile component
 - B. The less volatile component
 - C. Any one arbitrarily
 - D. The more viscous component
23. Which of the following is NOT a solution?
- A. Sugar dissolved in water
 - B. Sand suspended in water
 - C. Copper dissolved in gold
 - D. Alcohol mixed with water
24. Solutions containing only one solute and one solvent are called
- A. Binary solutions
 - B. Ternary solutions

- C. Multicomponent solutions
- D. Colloidal solutions

25. A true solution cannot be separated from solvent by

- A. Filtration
- B. Distillation
- C. Evaporation
- D. Crystallization

SECTION 2: CONCENTRATION UNITS – CONCEPTUAL MCQs (Q26–Q60)

26. Mass % of a component is defined as

- A. Mass of solute per 100 mL solution
- B. Mass of solute per 100 g solution
- C. Mass of solute per 1 L solution
- D. Mass of solute per 1 kg solvent

27. Mole fraction of component A in a binary solution is

- A. $\frac{n_A}{n_A + n_B}$
- B. $\frac{n_A}{n_B}$
- C. $\frac{n_B}{n_A + n_B}$
- D. $\frac{m_A}{m_A + m_B}$

28. Molarity is defined as

- A. Moles of solute per kg of solvent
- B. Moles of solute per litre of solution
- C. Moles of solvent per litre of solution
- D. Grams of solute per litre of solution

29. Molality is defined as

- A. Moles of solute per litre of solution
- B. Moles of solute per kg of solvent
- C. Moles of solvent per kg of solute
- D. Grams of solute per kg of solvent

30. The SI unit of molarity is
- A. mol kg^{-1}
 - B. mol L^{-1}
 - C. g L^{-1}
 - D. dimensionless
31. The SI unit of molality is
- A. mol kg^{-1}
 - B. mol L^{-1}
 - C. g L^{-1}
 - D. dimensionless
32. Which of the following has no unit?
- A. Molarity
 - B. Molality
 - C. Mole fraction
 - D. Mass %
33. Which of the following is temperature-dependent?
- A. Mole fraction only
 - B. Molarity only
 - C. Molality only
 - D. Both molarity and molality
34. Which is generally preferred for expressing concentration in colligative property calculations?
- A. Molarity
 - B. Molality
 - C. Volume %
 - D. Normality
35. Which concentration term is most suitable when temperature varies?
- A. Molarity
 - B. Molality
 - C. Mass %
 - D. ppm
36. The relation between moles, molarity and volume is
- A. $n = MV$

B. $M = nV$

C. $V = nM$

D. $M = \frac{V}{n}$

37. A 1 molal solution contains

- A. 1 mol solute in 1 L solvent
- B. 1 mol solute in 1 kg solvent
- C. 1 mol solute in 1 kg solution
- D. 1 mol solute in 1000 mL solution

38. A 1 M solution contains

- A. 1 mol solute in 1 kg solvent
- B. 1 mol solute in 1 L solvent
- C. 1 mol solute in 1 L solution
- D. 1 mol solute in 1 kg solution

39. Increasing temperature will generally

- A. Increase molarity
- B. Decrease molarity
- C. Increase molality
- D. Decrease molality

40. A solution labelled "0.9% NaCl (w/v)" means

- A. 0.9 g NaCl per 100 g solution
- B. 0.9 g NaCl per 100 mL solution
- C. 9 g NaCl per 100 mL solution
- D. 9 g NaCl per 100 g solution

41. For very dilute aqueous solution, density $\approx 1 \text{ g mL}^{-1}$, then numerically

- A. $M \approx m$
- B. $M > m$ always
- C. $M < m$ always
- D. M is unrelated to m

42. ppm is usually used for

- A. Concentrated solutions only
- B. Very dilute solutions

- C. Only gaseous solutions
 - D. Only solid solutions
43. 10% (w/w) NaCl solution means
- A. 10 g NaCl in 100 g solvent
 - B. 10 g NaCl in 100 g solution
 - C. 10 g NaCl in 1 L solution
 - D. 10 g NaCl in 100 mL solution
44. In a binary solution, if mole fraction of solute is 0.2, then mole fraction of solvent is
- A. 0.1
 - B. 0.2
 - C. 0.8
 - D. 1.2
45. The sum of mole fractions of all components in a solution is always
- A. 0
 - B. 1
 - C. >1
 - D. <1
46. Which concentration term uses mass of solvent instead of mass of solution?
- A. Mass %
 - B. Molarity
 - C. Molality
 - D. Volume %
47. For acid–base titration calculations, which concentration unit is often convenient?
- A. ppm
 - B. Normality
 - C. Molality
 - D. Mole fraction
48. The concentration of ions in drinking water is usually expressed in
- A. M
 - B. m
 - C. %
 - D. ppm

49. Which of the following is correct for 0.5 M NaCl solution?
- A. 0.5 mol NaCl per kg water
 - B. 0.5 mol NaCl per L solution
 - C. 0.5 mol NaCl per 100 g solution
 - D. 0.5 mol NaCl per 100 mL water
50. For solutions used in osmotic pressure experiments, which concentration is most naturally used?
- A. Mole fraction
 - B. Molality
 - C. Molarity
 - D. ppm
51. Which of these is NOT a way to express concentration?
- A. Mass percentage
 - B. Volume percentage
 - C. Atom percentage
 - D. Mole fraction
52. For a solution of glucose in water, which statement is true?
- A. Mole fraction of glucose + mole fraction of water = 0
 - B. Sum of mole fractions = 1
 - C. Mole fraction of glucose = mole fraction of water
 - D. Mole fraction of glucose = molality
53. To prepare 250 mL of 0.1 M HCl from 1 M HCl, volume of 1 M HCl required is
- A. 25 mL
 - B. 50 mL
 - C. 100 mL
 - D. 250 mL
54. Which of these is most accurate when solution density is not known?
- A. Molarity
 - B. Molality
 - C. % (v/v)
 - D. Normality
55. For reaction stoichiometry based on volume of solution, the most convenient unit is
- A. Molality

- B. Molarity
 - C. ppm
 - D. Mole fraction
56. The unit " g L^{-1} " directly tells
- A. Mass %
 - B. Moles per litre
 - C. Mass per litre of solution
 - D. Mass per litre of solvent
57. A solution has 0.02 mol of solute in 0.5 kg of solvent. Molality is
- A. 0.04 m
 - B. 0.02 m
 - C. 0.4 m
 - D. 0.1 m
58. 0.5 mol of NaCl is dissolved in enough water to make 250 mL of solution. Molarity is
- A. 0.25 M
 - B. 0.5 M
 - C. 1.0 M
 - D. 2.0 M
59. Which concentration term is independent of pressure?
- A. Molarity
 - B. Normality
 - C. Molality
 - D. Mass %
60. Which concentration term is exactly equal to mole fraction $\times 100$?
- A. Mole percent
 - B. Mass percent
 - C. Volume percent
 - D. ppm

SECTION 3: SOLUBILITY & HENRY'S LAW (Q61–Q90)

61. Solubility of a solute is
- A. Maximum amount of solute dissolved per unit mass of solvent at given T
 - B. Minimum amount of solute that dissolves at any T
 - C. Amount of solute per litre of solution
 - D. Always independent of temperature
62. Solubility of most solid solutes in liquids generally
- A. Decreases with increase in temperature
 - B. Increases with increase in temperature
 - C. Does not change
 - D. Becomes zero
63. Solubility of gases in liquids generally
- A. Increases with rise in temperature
 - B. Decreases with rise in temperature
 - C. Is independent of pressure
 - D. Is very high at all temperatures
64. Henry's law relates
- A. Solubility of solid to temperature
 - B. Solubility of gas to partial pressure
 - C. Vapour pressure to temperature
 - D. Boiling point to pressure
65. According to Henry's law, at constant temperature
- A. $p = k_H c$
 - B. $c = \frac{k_H}{p}$
 - C. $p = \frac{c}{k_H}$
 - D. None
66. In Henry's law, p denotes
- A. Osmotic pressure
 - B. Partial pressure of gas above solution
 - C. Vapour pressure of pure solvent
 - D. Total pressure of solution

67. In Henry's law, c is
- A. Concentration of solvent
 - B. Concentration of gas in solution
 - C. Concentration of all solutes
 - D. Concentration of solid
68. A higher value of Henry's constant k_H for a gas in a solvent means
- A. Higher solubility
 - B. Lower solubility
 - C. No relation
 - D. Infinite solubility
69. At a given temperature, doubling the pressure of a gas over solution approximately
- A. Doubles its solubility
 - B. Halves its solubility
 - C. Makes no change
 - D. Makes it insoluble
70. Which gas obeys Henry's law most closely in water?
- A. NH_3
 - B. CO_2
 - C. O_2
 - D. HCl
71. Henry's law is more applicable to
- A. Highly soluble gases only
 - B. Slightly soluble gases at low pressure
 - C. Only polar gases
 - D. Only non-polar solvents
72. The unit of Henry's constant can be
- A. mol kg^{-1}
 - B. Pa (or atm)
 - C. $\text{Pa m}^3 \text{mol}^{-1}$
 - D. Both B and C
73. For soft drinks, CO_2 gas is filled under
- A. Low pressure, high T

- B. High pressure, low T
 - C. High pressure, high T
 - D. Low pressure, low T
74. "Bends" faced by deep-sea divers are related to
- A. Henry's law for O_2
 - B. Henry's law for N_2
 - C. Solubility of CO_2 only
 - D. Vapour pressure of water
75. Which of the following increases solubility of a gas in a liquid?
- A. Increase in temperature
 - B. Decrease in pressure
 - C. Increase in pressure
 - D. Both A and B
76. Why does aerated water taste better when cold?
- A. Gas solubility is less at low T
 - B. Gas solubility is more at low T
 - C. Vapour pressure is maximum at low T
 - D. No relation
77. Solubility of gases in liquids is expressed in
- A. $g\ L^{-1}$ or $mol\ L^{-1}$
 - B. $mol\ kg^{-1}$
 - C. % by mass
 - D. ppm only
78. Which of the following is NOT a factor affecting solubility of gas in liquid?
- A. Temperature
 - B. Pressure
 - C. Nature of gas
 - D. Colour of gas
79. Solubility of O_2 in blood is crucially governed by
- A. Raoult's law
 - B. Henry's law

- C. Dalton's law
- D. Boyle's law

80. When a sealed soda bottle is opened at room temperature

- A. Pressure above liquid decreases and gas escapes
- B. Pressure increases
- C. Solubility of gas increases
- D. None

81. Addition of an inert gas to the vapour space (keeping partial pressure of the gas same)

- A. Changes solubility according to Henry's law
- B. Does not change solubility
- C. Doubles solubility
- D. Converts solution to ideal

82. Henry's law is analogous to

- A. Raoult's law for volatile solvent
- B. Dalton's law for gases
- C. Boyle's law for gases
- D. Avogadro's law

83. Which is a limitation of Henry's law?

- A. Only valid at very high pressures
- B. Only valid when gas reacts with solvent
- C. Deviations occur at high gas pressures or non-ideal behaviour
- D. It cannot be used for real gases

84. For a gas obeying Henry's law, solubility vs pressure graph is

- A. Straight line passing through origin
- B. Parabola
- C. Horizontal line
- D. Vertical line

85. Increase in temperature usually

- A. Decreases Henry's constant
- B. Increases Henry's constant
- C. Eliminates Henry's law
- D. Has no effect

86. Which one is TRUE for solubility of solids?

- A. Always increases with T
- B. Always decreases with T
- C. May increase or decrease with T
- D. Is independent of T

SECTION 4: RAOULT'S LAW, IDEAL & NON-IDEAL SOLUTIONS (Q91–Q135)

91. Raoult's law for a solution of volatile liquid A and non-volatile solute states that vapour pressure of solvent is

- A. Inversely proportional to its mole fraction
- B. Directly proportional to its mole fraction
- C. Independent of composition
- D. Equal to vapour pressure of pure solvent

92. For an ideal solution of volatile liquids A and B, partial vapour pressure of A is

- A. $p_A = x_A p_A^\circ$
- B. $p_A = p_A^\circ$
- C. $p_A = x_B p_A^\circ$
- D. $p_A = x_A p_B^\circ$

93. In a binary ideal solution, total vapour pressure is

- A. Sum of partial pressures
- B. Product of partial pressures
- C. Average of pure vapour pressures
- D. Independent of composition

94. An ideal solution obeys Raoult's law

- A. At all compositions and T
- B. Only at infinite dilution
- C. Only at high pressure
- D. Never exactly

95. In an ideal solution, enthalpy of mixing ΔH_{mix} is

- A. Positive
- B. Negative

- C. Zero
- D. Extremely large

96. In an ideal solution, volume change on mixing ΔV_{mix} is

- A. Zero
- B. Positive
- C. Negative
- D. Infinite

97. Positive deviation from Raoult's law arises when

- A. A-B interactions stronger than A-A and B-B
- B. A-B interactions weaker than A-A and B-B
- C. Forces are identical
- D. None

98. Negative deviation from Raoult's law arises when

- A. A-B interactions weaker
- B. A-B interactions stronger
- C. No new interactions
- D. Forces negligible

99. A solution with higher vapour pressure than predicted by Raoult's law shows

- A. Positive deviation
- B. Negative deviation
- C. No deviation
- D. Azeotropic behaviour

100. A solution with lower vapour pressure than predicted by Raoult's law shows

- A. Positive deviation
- B. Negative deviation
- C. Ideal behaviour
- D. No mixing

101. Which of the following is nearly ideal?

- A. Ethanol-water
- B. Acetone-chloroform
- C. Benzene-toluene
- D. HCl-water

102. Ethanol–water mixture shows
- A. Ideal behaviour
 - B. Positive deviation
 - C. Negative deviation
 - D. No mixing
103. Acetone–chloroform shows
- A. Ideal behaviour
 - B. Positive deviation
 - C. Negative deviation
 - D. Immiscibility
104. In solutions showing positive deviation, boiling point is
- A. Higher than either pure component
 - B. Lower than both pure components
 - C. Equal to average
 - D. Independent
105. In solutions showing negative deviation, boiling point is
- A. Lower than either pure component
 - B. Higher than both pure components
 - C. Equal to average
 - D. Independent
106. Azeotropes are
- A. Only ideal solutions
 - B. Constant boiling mixtures
 - C. Always immiscible
 - D. Always solid solutions
107. Minimum boiling azeotrope corresponds to
- A. Positive deviation
 - B. Negative deviation
 - C. Ideal behaviour
 - D. Non-volatile solute
108. Maximum boiling azeotrope corresponds to
- A. Positive deviation

- B. Negative deviation
 - C. Ideal behaviour
 - D. Ideal gas
109. Example of minimum boiling azeotrope is
- A. HCl–water
 - B. Ethanol–water
 - C. Nitric acid–water
 - D. Glycerol–water
110. Example of maximum boiling azeotrope is
- A. Ethanol–water
 - B. Acetone–CCl₄
 - C. HCl–water or nitric acid–water (depending on system)
 - D. Benzene–toluene
111. Raoult's law can be viewed as a special case of Henry's law when
- A. $k_H = p^\circ$ and solute is similar to solvent
 - B. $k_H = 0$
 - C. $k_H \rightarrow \infty$
 - D. Gas reacts with solvent
112. For an ideal binary solution, total vapour pressure vs mole fraction curve is
- A. Straight line
 - B. Parabola
 - C. Horizontal line
 - D. Vertical line
113. Dalton's law of partial pressure states that for non-reacting gases
- A. Total pressure = product of partial pressures
 - B. Total pressure = sum of partial pressures
 - C. Total pressure = average pressure
 - D. Total pressure = difference of partial pressures
114. For a binary ideal solution of volatile liquids, Raoult's law is applied to
- A. Solute only
 - B. Solvent only

- C. Both components
- D. Neither

115. Which of the following is NOT a property of ideal solution?

- A. $\Delta H_{\text{mix}} = 0$
- B. $\Delta V_{\text{mix}} = 0$
- C. Obeys Raoult's law at all compositions
- D. Strong chemical interaction between components

SECTION 5: COLLIGATIVE PROPERTIES – BASIC MCQs (Q136–Q200)

(Only key pattern; you can ask explicitly for full SOLUTIONS_LEVEL2 extension if needed.)

116. Colligative properties depend on

- A. Nature of solute only
- B. Number of solute particles only
- C. Nature and number both
- D. Solvent only

117. Which is NOT a colligative property?

- A. Relative lowering of vapour pressure
- B. Elevation in boiling point
- C. Depression in freezing point
- D. Viscosity

118. Relative lowering of vapour pressure is

- A. $\frac{p^\circ - p}{p^\circ}$
- B. $\frac{p}{p^\circ}$
- C. $\frac{p^\circ}{p}$
- D. $\frac{p^\circ - p}{p}$

119. Elevation in boiling point ΔT_b is proportional to

- A. Molarity
- B. Molality

C. Mole fraction

D. Volume %

120. Depression in freezing point ΔT_f is proportional to

A. Molality

B. Normality

C. ppm

D. Mass %

121. Osmotic pressure is

A. Pressure required to stop osmosis

B. Pressure needed to start osmosis

C. Always zero

D. Difference in vapour pressure

122. Van't Hoff factor i is

A. $\frac{\text{number of particles after}}{\text{number before}}$

B. $\frac{\text{measured colligative property}}{\text{calculated for nonelectrolyte}}$

C. Both A and B (equivalent)

D. Always 1

123. For non-electrolyte solutes, i is usually

A. 0

B. 1

C. 2

D. 3

124. For NaCl (assumed complete dissociation) in water, i is approximately

A. 1

B. 2

C. 0.5

D. 3

125. Presence of non-volatile solute in solvent

A. Increases vapour pressure

B. Decreases vapour pressure

- C. Does not change vapour pressure
- D. First increases then decreases

126. The equation for osmotic pressure of ideal solution is

- A. $\pi = CRT$
- B. $\pi = nRT$
- C. $\pi = \frac{RT}{c}$
- D. $\pi = kT$

127. Which method is most suitable to determine molar mass of large biomolecules?

- A. Boiling point elevation
- B. Freezing point depression
- C. Osmotic pressure
- D. Vapour pressure lowering

128. Isotonic solutions have

- A. Same solute
- B. Same solvent
- C. Same osmotic pressure
- D. Same viscosity

129. Hypertonic solution relative to cell fluid has

- A. Lower osmotic pressure
- B. Higher osmotic pressure
- C. Same osmotic pressure
- D. No osmotic pressure

130. Hypotonic solution relative to cell fluid has

- A. Higher osmotic pressure
- B. Lower osmotic pressure
- C. Same osmotic pressure
- D. Negative osmotic pressure

131. Depression in freezing point is given by

- A. $\Delta T_f = K_f m$
- B. $\Delta T_f = K_b m$

C. $\Delta T_f = Km$ (no constant)

D. $\Delta T_f = Km^2$

132. Elevation in boiling point is given by

A. $\Delta T_b = K_b m$

B. $\Delta T_b = K_f m$

C. $\Delta T_b = Km^2$

D. $\Delta T_b = Km$

133. Molal depression constant depends on

A. Nature of solute

B. Nature of solvent

C. Both solute and solvent

D. Concentration

134. For a given solvent, K_f is

A. Constant

B. Variable

C. Zero

D. Infinite

135. If measured colligative property is greater than expected, solute is likely

A. Associating

B. Dissociating

C. Non-electrolyte

D. None

136. If measured colligative property is smaller than expected, solute is likely

A. Associating

B. Dissociating

C. Neutral

D. Ideal gas

137. Which colligative property is easiest to measure accurately for very dilute solutions?

A. Rt. lowering of vapour pressure

B. Elevation in boiling point

C. Depression in freezing point

D. Osmotic pressure

138. When water freezes at 0 °C, presence of solute causes
- A. Increase in freezing point
 - B. Decrease in freezing point
 - C. No change
 - D. Variable effect
139. Antifreeze solutions in car radiators work mainly because of
- A. Increased boiling point
 - B. Decreased freezing point
 - C. Both A and B
 - D. Lower vapour pressure
140. Colligative properties are valid mainly for
- A. Concentrated solutions
 - B. Dilute solutions
 - C. Suspensions
 - D. Colloids
141. For 1 m glucose and 1 m NaCl solution (ideal dissociation), which has higher boiling point?
- A. Glucose
 - B. NaCl
 - C. Same
 - D. Cannot say
142. Which property can distinguish between glucose and NaCl solution of same molality?
- A. Colour
 - B. Odour
 - C. Colligative property
 - D. Density only
143. If van't Hoff factor $i = 3$ for solute AB_2 , its degree of dissociation α (complete formula) is
- A. 1
 - B. $\frac{2}{3}$
 - C. $\frac{i-1}{2}$
 - D. $\frac{3-1}{2}$

144. Osmotic pressure has same units as
- A. Temperature
 - B. Pressure
 - C. Concentration
 - D. Volume
145. An ideal semipermeable membrane allows passage of
- A. Solvent only
 - B. Solute only
 - C. Both equally
 - D. Neither
146. Reverse osmosis involves
- A. Applying pressure less than osmotic pressure
 - B. Applying pressure greater than osmotic pressure
 - C. Heating solution
 - D. Cooling solution
147. In water purification by reverse osmosis, which passes through membrane?
- A. Ions and impurities
 - B. Solvent water
 - C. Both
 - D. Neither
148. If osmotic pressure of 0.1 M urea solution at 300 K is π , osmotic pressure of 0.2 M urea solution at same T is approx.
- A. $\pi/2$
 - B. π
 - C. 2π
 - D. 4π
149. If two solutions have same osmotic pressure, they are said to be
- A. Hypotonic
 - B. Hypertonic
 - C. Isotonic
 - D. Colloidal

150. Colligative properties depend on

- A. Moles of solute
- B. Mass of solute
- C. Type of solute
- D. Shape of solute

151–200: (More numerical MCQs on Raoult's law and colligative properties can be added similarly on request: dilution calculations, mole-fraction based Raoult's law sums, simple elevation in boiling point / depression in freezing point questions, and basic van't Hoff factor numericals.)